FLAT FINITE TYPE ALGEBRAS AND FLAT FINITE MODULES OVER A VALUATION RING ARE FINITELY PRESENTED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness over a valuation ring is the same as torsion freeness. Lemma \ref{lemma-valuation-ring-torsion-free-flat}, cited at proof lines 5981-5982. Lets the author remove torsion parts, which is the key reduction.
- Homogenisation: a finite type algebra over a ring is the degree zero part of a graded algebra generated in degree one, and similarly for modules. Algebra, Lemma \ref{algebra-lemma-homogenize}, cited at proof lines 5983-5988. Realises the algebra and the module as localisations of graded objects.
- The torsion submodule can be removed without changing the degree zero part of a graded module. Lemma \ref{lemma-torsion}, cited at proof lines 5989-5992. Makes both the graded algebra and the graded module torsion free.
- A flat graded algebra of finite type over a local ring and a flat finite graded module over it are finitely presented. Lemma \ref{lemma-flat-graded-finite-type-finite-presentation} and Lemma \ref{lemma-flat-graded-finite-type-finite-presentation-module}, cited at proof lines 5994-5996 and 5999-6001. Give the two graded statements which specialise to the required results.

Source reference. Original lemma 053E: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6602–6602. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0539: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6597–6597. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 052N: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6576–6576. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0537: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6595–6595. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 053D: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6601–6601. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 053C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6600–6600. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original reference cite-Nagata-Finitely: sources/stacks-more-algebra/pinned-original-source.tex, source lines 5966–5966. Bibliographic pointer appearing in the counted text; it carries no mathematical argument.

Lemma 0.2. *Let A be a valuation ring. Let $A \to B$ be a ring map of finite type. Let M be a finite B-module.*

(1) *If B is flat over A, then B is a finitely presented A-algebra.*
(2) *If M is flat as an A-module, then M is finitely presented as a B-module.*

Proof. We are going to use that an A-module is flat if and only if it is torsion free, see Lemma 0539. By Algebra, Lemma 052N we can find a graded A-algebra S with $S_0 = A$ and generated by finitely many elements in degree 1, an element $f \in S_1$ and a finite graded S-module N such that $B \cong S_{(f)}$ and $M \cong N_{(f)}$. If M is torsion free, then we can take N torsion free by replacing it by N/N_{tors}, see Lemma 0537. Similarly, if B is torsion free, then we can take S torsion free by replacing it by S/S_{tors}. Hence in case (1), we may apply Lemma 053D to see that S is a finitely presented A-algebra, which implies that $B = S_{(f)}$ is a finitely presented A-algebra. To see (2) we may first replace S by a graded polynomial ring, and then we may apply Lemma 053C to conclude. □